GLOBAL IMPACT OF AI CONTENT

This dataset provides an in-depth view of how artificial intelligence (AI) is impacting different countries and industries over time. It contains data on AI adoption, job displacement, revenue changes, human-AI collaboration, consumer trust, and market share, offering a multidimensional analysis of AI's global footprint.

Dataset link:
https://www.kaggle.com/code/muhammedaliyilmazz/global-ai-content-impact-dataset-data-analysis

Data Sources: Public surveys, industry reports, AI research publications and digital media analytics

Query 1

```sql
SELECT
    Industry,
    ROUND(AVG(CAST(`Revenue Increase Due to AI (%)` AS FLOAT)), 2) AS avg_revenue_increase,
    ROUND(AVG(CAST(`Job Loss Due to AI (%)` AS FLOAT)), 2)         AS avg_job_loss,
    ROUND(
        AVG(CAST(`Revenue Increase Due to AI (%)` AS FLOAT)) /
        NULLIF(AVG(CAST(`Job Loss Due to AI (%)` AS FLOAT)), 0),
    2) AS roi_per_job_loss
FROM ai
GROUP BY Industry
ORDER BY roi_per_job_loss DESC;
```

Result Grid

Industry	avg_revenue_increase	avg_job_loss	roi_per_job_loss
Media	43.72	22.75	1.92
Marketing	36.81	19.58	1.88
Retail	37.68	21.85	1.72
Automotive	46.48	28.92	1.61
Education	39.54	26.14	1.51
Healthcare	38.59	25.58	1.51
Legal	41.82	28.23	1.48
Manufacturing	42.81	32.75	1.31
Finance	36.26	27.79	1.3
Gaming	33.23	27.2	1.22

1. Industry Efficiency: ROI per Job Loss

This query calculates a Human-to-Revenue ratio: for every percentage of job loss, how much revenue growth is actually achieved?

- **Insight 1: The Creative Sweet Spot (Media & Marketing).** Media (1.92) and Marketing (1.88) have the highest ROI ratios. This suggests that in these fields, AI isn't just a replacement for people; it’s a massive force multiplier. A relatively small

reduction in headcount (or shift in roles) results in nearly double the revenue growth.

  - **Action that can be taken:** Transition from Generalist roles to **AI Orchestrator** roles. Since the ROI is so high here, companies should reinvest those gains into high-level creative strategy that AI can't replicate yet.

- **Insight 2: Manufacturing's Brutal Math.** Manufacturing has the highest job loss (32.75) but one of the lower ROI ratios (1.31). This implies that AI in manufacturing is currently being used for **low-margin automation**, replacing expensive human labor to save costs rather than creating explosive new revenue streams.

  - **Action that can be taken:** Shift focus from **Labor Replacement to Product Innovation.** To raise that ratio, manufacturers should use AI for generative design (creating better products) rather than just robotic assembly (making the same products cheaper).

```sql
SELECT
    Country,
    `Regulation Status`,
    ROUND(AVG(`Revenue Increase Due to AI (%)`), 2) AS avg_revenue_increase,
    ROUND(AVG(`Consumer Trust in AI (%)`), 2) AS avg_trust,
    ROUND(AVG(`AI Adoption Rate (%)`), 2) AS avg_adoption
FROM ai
GROUP BY Country, `Regulation Status`
ORDER BY Country ASC, `Regulation Status` ASC, avg_revenue_increase DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Country	Regulation Status	avg_revenue_increase	avg_trust	avg_adoption
Australia	Lenient	56.09	64.58	44.13
Australia	Moderate	33.7	71.13	49.37
Australia	Strict	58.39	57.75	66
Canada	Lenient	38.44	62.66	49.62
Canada	Moderate	25.97	70.98	52.31
Canada	Strict	45.51	57.14	55.12
China	Lenient	54.75	62.64	38.86
China	Moderate	38.97	58.28	59.27
China	Strict	35.87	61.06	57.35
France	Lenient	50.68	67.07	55.6
France	Moderate	43.53	57.77	56.61

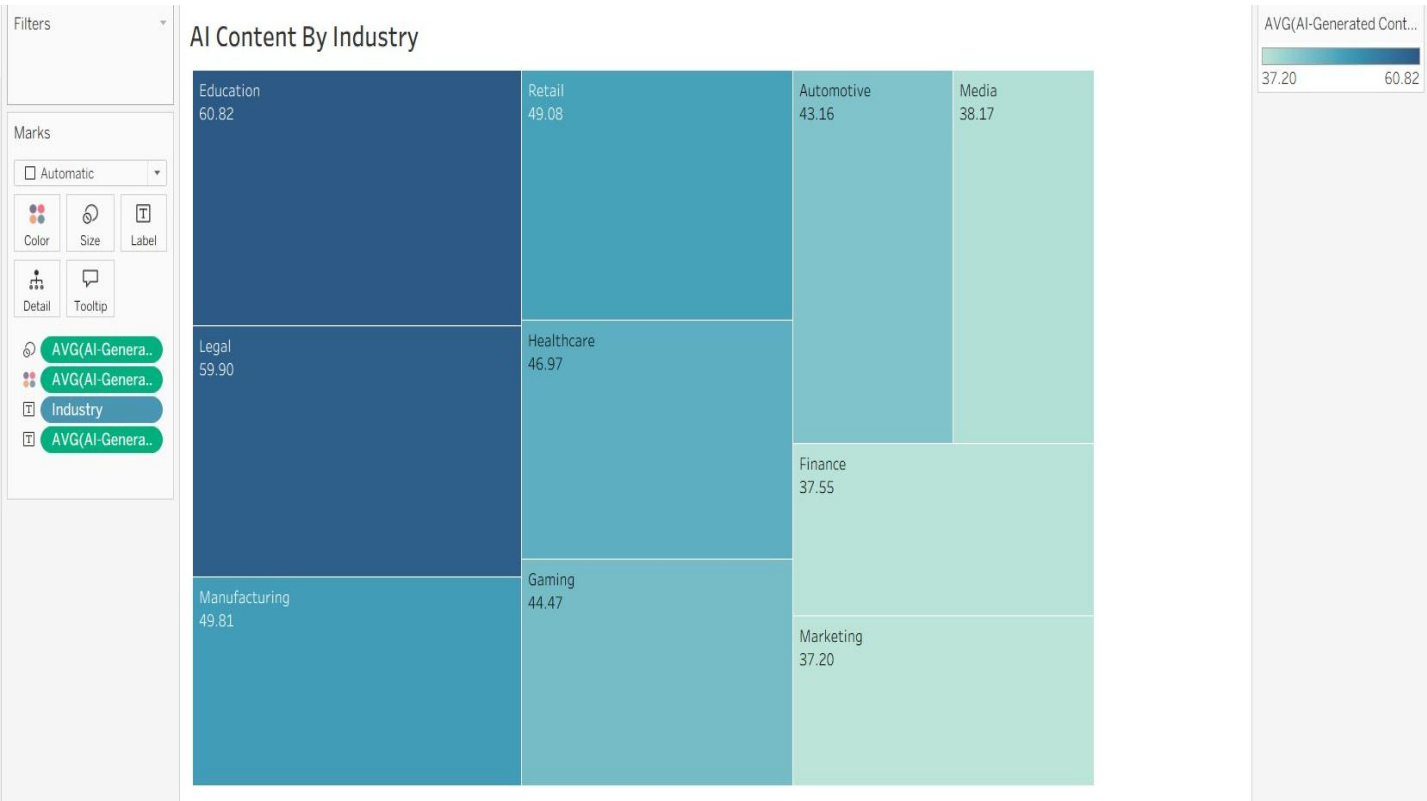
2. Regulation Status vs. Economic Impact

This query looks at how different rules of the road (Lenient, Moderate, Strict) affect trust, adoption and revenue.

- **Insight 1: The Moderate Trust Peak.** In almost every country (Australia, Canada, etc.), **Moderate Regulation** leads to the highest Consumer Trust (71.13% in Australia). However, it often results in the *lowest* revenue increase. It seems playing it safe makes people happy but slows down the money-making machine.
    - **Action:** Use Moderate frameworks as a **marketing and PR tool.** If your brand relies on consumer sentiment, aligning with moderate regulatory standards is your best bet for building long-term brand equity.
- **Insight 2: The Strict Regulation Paradox.** Surprisingly, in Australia and Canada, **Strict Regulation** actually correlates with the *highest* AI Adoption Rates (66% and 55.12% respectively). When the rules are clear, even if they are tough, businesses feel more confident investing because they aren't afraid of a sudden, unpredictable legal crackdown.

    - **Action:** Push for **Regulatory Clarity.** Businesses shouldn't lobby for *less* regulation, but for *clearer* regulation. The data suggests that a Strict but known environment is better for adoption than a Lenient but vague one.

- **Insight 3: China's Lenient Revenue Engine.** China sees a massive revenue jump (54.75 %) under Lenient regulation, but trust is significantly lower than in other regions. This is a Growth at all Costs model.
    - **Action:** International firms operating in lenient zones should **adopt Strict internal ethics** to remain competitive in global markets where trust is a currency.

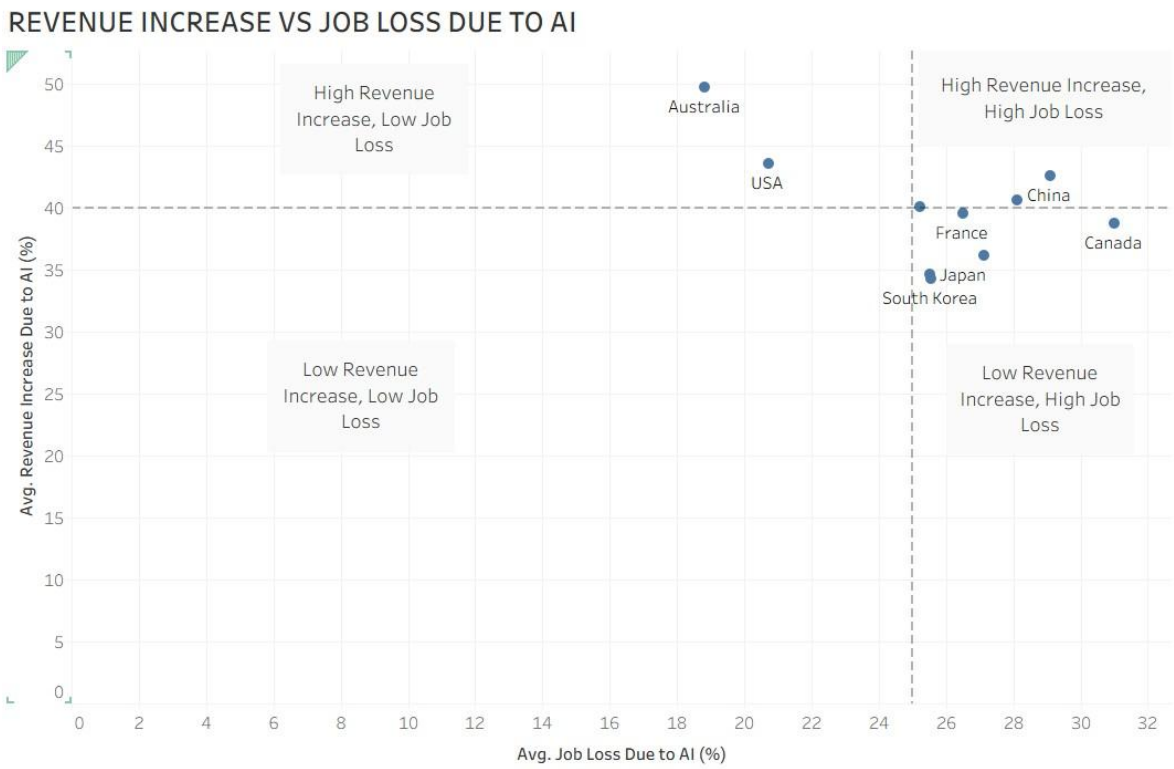


1. AI Content By Industry (Treemap)

This chart illustrates the average volume of AI-generated content across various sectors.

- **Insight 1: Education and Legal lead the pack.** These two industries show significantly higher AI content volumes (60.82 TB and 59.90 TB respectively) compared to others. This suggests a heavy reliance on AI for drafting documents, research and educational materials.
    - **Action:** Implement **robust verification and Human-in-the-Loop protocols** in these sectors. Because the stakes are high (legal accuracy and student learning), automated content must be audited for hallucinations to prevent misinformation.
- **Insight 2: Marketing and Media show surprisingly lower volume.** Despite being the face of generative AI, these sectors have the lowest volumes (37.20 TB and 38.17 TB). This might indicate a preference for high-quality, human-curated content over mass-produced AI output.

- **Action:** Focus on **AI-augmentation rather than automation**. For these industries, the goal should be using AI to brainstorm or A/B test ideas rather than generating the final public-facing output.



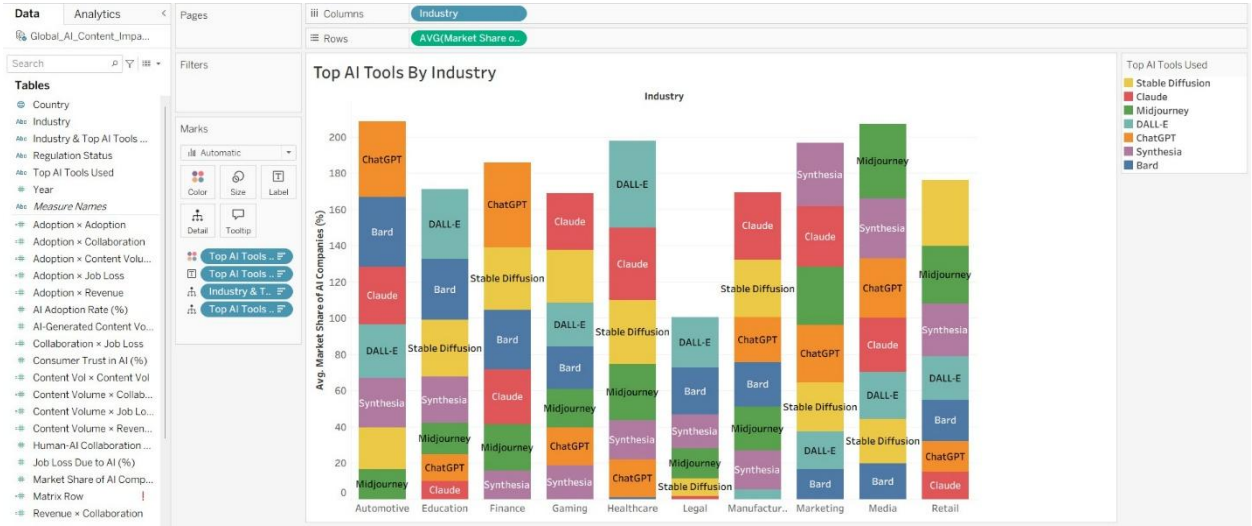
2. Revenue Increase vs. Job Loss Due to AI (Scatter Plot)

This visualization plots economic gains against human labor displacement.

- **Insight 1: The Efficiency Leaders (USA and Australia).** These countries sit in the high-revenue, relatively lower job-loss quadrant. They are successfully turning AI into a force multiplier rather than a simple replacement for humans.

- **Action:** Analyze the **reskilling models** used in these countries. Other nations should study their tax incentives or corporate training programs that encourage workers to use AI tools to increase their output.

- **Insight 2: The Risk Zone (Japan, South Korea, and Canada).** These countries are seeing high job loss percentages (>25%) but relatively lower revenue gains compared to the USA. This suggests AI is being used primarily for cost-cutting (layoffs) rather than innovation.
    - **Action:** Governments in these regions should prioritize **AI-driven job transition programs**. If the revenue isn't offsetting the social cost of job loss, policy intervention may be needed to steer AI use toward new product creation rather than just head-count reduction.



3. Top AI Tools By Industry (Stacked Bar Chart)

This chart displays the market share of specific AI tools (ChatGPT, Claude, Bard, Midjourney, etc.) across industries.

- **Insight 1: Media and Gaming are Visual-First.** Midjourney and DALL-E have massive footprints in Media and Gaming, while ChatGPT takes a smaller slice. These industries have clearly found their power tools for asset generation.

    - **Action:** Invest in **Intellectual Property (IP) protection tools**. As visual AI dominates these sectors, companies need clear legal frameworks and watermarking technologies to protect their original styles from being absorbed into training sets.

- **Insight 2: Tool Fragmentation in Retail and Manufacturing.** Unlike Finance (where ChatGPT and Bard dominate), Retail uses a very even mix of almost every tool, including Synthesia and Stable Diffusion.
    - **Action:** Prioritize **interoperability and API integration**. Since Retailers are using a cocktail of different AI models, they should focus on building platforms that allow these different tools to talk to each other to avoid data silos.

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